

1 Gain law

(bars denote normalized quantities: lengths by R , $\bar{a} = a/a_o$, $\bar{f} = a_o f$; $0 < \bar{a} < 1$; $e_\theta = \theta - \hat{\theta}$)

$$\bar{a}_{\text{true}}(\bar{w}) = 1 - \frac{9}{8\bar{w}} + \frac{1}{2\bar{w}^3} - \frac{57}{100\bar{w}^4} + \frac{1}{5\bar{w}^5} + \frac{7}{200\bar{w}^{11}} - \frac{1}{25\bar{w}^{12}}$$

$$\bar{a}'_{\text{true}}(\bar{w}) > 0 \quad \text{on} \quad \bar{w} > 1 \quad \implies \quad \bar{a}' = b_{\text{true}}(\bar{a})(1 - \bar{a})^2 \quad \text{exactly,} \quad b_{\text{true}} := \frac{\bar{a}'_{\text{true}}}{(1 - \bar{a}_{\text{true}})^2}$$

$$\begin{aligned} \bar{a}_{\text{true}}(1) = 0, \quad \bar{a}'_{\text{true}}(1) = 1 &\implies b_{\text{true}}(\bar{a} = 0) = 1 \\ 1 - \bar{a}_{\text{true}} = \frac{9}{8\bar{w}} + \mathcal{O}(\bar{w}^{-3}) &\implies b_{\text{true}}(\bar{a} \rightarrow 1) = \frac{8}{9} \end{aligned}$$

$$\begin{aligned} \bar{a}'(\bar{a}, b) &:= b(1 - \bar{a})^2 \\ a_w &= a_o \bar{a}_w, \quad a_o = \frac{\Delta t}{\gamma_N R} \end{aligned}$$

2 Jacobians

$$\begin{aligned} \frac{\partial \bar{a}'}{\partial \bar{a}} &= -2b(1 - \bar{a}) \\ \frac{\partial \bar{a}'}{\partial b} &= (1 - \bar{a})^2 = \frac{\bar{a}'}{b} \end{aligned}$$

3 True dynamics

$$(\delta \bar{w}_j[k] := \delta \bar{w}[k - d + j - 1], \quad j = 1, \dots, d + 1, \quad d = 2)$$

$$\begin{aligned} \bar{w}[k] &= \bar{w}_d[k] - \delta \bar{w}_3[k] \\ \bar{w}_d[k + 1] &= \bar{w}_d[k] + \Delta \bar{w}_d[k] \\ \delta \bar{w}_1[k + 1] &= \delta \bar{w}_2[k] \\ \delta \bar{w}_2[k + 1] &= \delta \bar{w}_3[k] \\ \delta \bar{w}_3[k + 1] &= \lambda_c \delta \bar{w}_3[k] - F_{dw}[k] e_{\bar{a}_w}[k] - \varepsilon_w[k] \\ \bar{a}_w[k + 1] &= \bar{a}_w[k] + \bar{a}'_{\text{true}}[k] \left[\Delta \bar{w}_d[k] + (1 - \lambda_c) \delta \bar{w}_3[k] + F_{dw}[k] e_{\bar{a}_w}[k] + \varepsilon_w[k] \right] \\ b_{\text{true}}[k] &= b_{\text{true}}(\bar{a}_w[k]) \end{aligned}$$

$$F_{dw}[k] = \bar{f}_{dw}[k] + (1 - \lambda_c) \sum_{i=1}^d \bar{f}_{dw}[k - i]$$

$$\varepsilon_w[k] = \bar{a}_w[k] \bar{f}_T[k] + (1 - \lambda_c) \sum_{i=1}^d \bar{a}_w[k - i] \bar{f}_T[k - i] + (1 - \lambda_c) n_w[k]$$

4 Process model

$$\mathbf{x} = [\delta\bar{w}_1, \delta\bar{w}_2, \delta\bar{w}_3, \bar{a}_w, b]^\top \quad (d = 2)$$

5 Estimator

$$\begin{aligned} \delta\hat{w}_1[k+1] &= \delta\hat{w}_2[k] \\ \delta\hat{w}_2[k+1] &= \delta\hat{w}_3[k] \\ \delta\hat{w}_3[k+1] &= \lambda_c \delta\hat{w}_3[k] \\ \hat{a}_w[k+1] &= \hat{a}_w[k] + \hat{b}[k] (1 - \hat{a}_w[k])^2 [\Delta\bar{w}_d[k] + (1 - \lambda_c) \delta\hat{w}_3[k]] \\ \hat{b}[k] &\equiv \hat{b}[0] \end{aligned}$$

6 Error dynamics

(True – Estimator; coefficients at the estimate)

$$\begin{aligned} e_{\delta\bar{w}_1}[k+1] &= e_{\delta\bar{w}_2}[k] \\ e_{\delta\bar{w}_2}[k+1] &= e_{\delta\bar{w}_3}[k] \\ e_{\delta\bar{w}_3}[k+1] &= \lambda_c e_{\delta\bar{w}_3}[k] - F_{dw}[k] e_{\bar{a}_w}[k] - \varepsilon_w[k] \\ e_{\bar{a}_w}[k+1] &= \left[1 + \hat{b}(1 - \hat{a}_w)^2 F_{dw} - 2\hat{b}(1 - \hat{a}_w)(\Delta\bar{w}_d + (1 - \lambda_c)\delta\hat{w}_3) \right] e_{\bar{a}_w}[k] \\ &\quad + (1 - \lambda_c) \hat{b}(1 - \hat{a}_w)^2 e_{\delta\bar{w}_3}[k] + (1 - \hat{a}_w)^2 (\Delta\bar{w}_d + (1 - \lambda_c)\delta\hat{w}_3) e_b[k] \\ &\quad + \hat{b}(1 - \hat{a}_w)^2 \varepsilon_w[k] \\ e_b[k] &= b_{\text{true}}(\bar{a}_w[k]) - \hat{b}[0], \quad |e_b[k]| \leq \sqrt{P_{bb}} \end{aligned}$$

$$F_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & \lambda_c & -F_{dw} & 0 \\ 0 & 0 & (1 - \lambda_c) \hat{b}(1 - \hat{a}_w)^2 & 1 + \hat{b}(1 - \hat{a}_w)^2 F_{dw} & (1 - \hat{a}_w)^2 \\ 0 & 0 & 0 & -2\hat{b}(1 - \hat{a}_w)(\Delta\bar{w}_d + (1 - \lambda_c)\delta\hat{w}_3) & \times (\Delta\bar{w}_d + (1 - \lambda_c)\delta\hat{w}_3) \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{q}[k] = \begin{bmatrix} 0, & 0, & -\varepsilon_w, & \hat{b}(1 - \hat{a}_w)^2 \varepsilon_w, & 0 \end{bmatrix}^\top$$

7 Measurement and innovation

$$\begin{aligned} \nabla_d \bar{w}_d[k] &= \bar{w}_d[k] - \bar{w}_d[k-d] \\ y_1[k] &= \delta \bar{w}_1[k] + n_w[k] \\ y_2[k] &= \bar{a}_w[k-d] + n_a[k] \end{aligned}$$

$$\bar{a}_w[k-d] = \bar{a}_w[k] - b(1 - \bar{a}_w[k])^2 \nabla_d \bar{w}_d[k]$$

$$\begin{aligned} \frac{\partial y_2}{\partial \bar{a}_w} &= 1 + 2\hat{b}(1 - \hat{a}_w) \nabla_d \bar{w}_d \\ \frac{\partial y_2}{\partial b} &= -(1 - \hat{a}_w)^2 \nabla_d \bar{w}_d \end{aligned}$$

$$H[k] = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 + 2\hat{b}(1 - \hat{a}_w) \nabla_d \bar{w}_d & -(1 - \hat{a}_w)^2 \nabla_d \bar{w}_d \end{bmatrix}$$

$$L[k] = P^f H^\top (H P^f H^\top + R)^{-1}, \quad L_{5,\cdot}[k] := 0$$

$$\begin{aligned} e_{y_1}[k] &= e_{\delta \bar{w}_1}[k] + n_w[k] \\ e_{y_2}[k] &= \left[1 + 2\hat{b}(1 - \hat{a}_w) \nabla_d \bar{w}_d \right] e_{\bar{a}_w}[k] - (1 - \hat{a}_w)^2 \nabla_d \bar{w}_d e_b[k] + n_a[k] \end{aligned}$$

8 Process and measurement noise

$$\begin{aligned} q_3 &= -\bar{a}_w \bar{f}_T \\ q_4 &= -\hat{b}(1 - \hat{a}_w)^2 q_3 \\ \gamma(\bar{w}) &= \gamma_N c(\bar{w}) \\ \sigma_{f_T}^2 &= a_o^2 \frac{4k_B T \gamma(\bar{w})}{\Delta t} \end{aligned}$$

$$\begin{aligned}
Q_{33} &= \frac{4k_B T a_o}{R} \bar{a}_w \\
Q_{34} = Q_{43} &= -\hat{b}(1 - \hat{a}_w)^2 Q_{33} \\
Q_{44} &= \hat{b}^2(1 - \hat{a}_w)^4 Q_{33} + q_{\text{law}}[k] \\
Q_{55} &= 0, \quad P_{55}[k] \equiv P_{bb}
\end{aligned}$$

$$\begin{aligned}
g[k] &:= F_e[k]_{45} = (1 - \hat{a}_w[k])^2 \left[\Delta \bar{w}_d[k] + (1 - \lambda_c) \delta \hat{w}_3[k] \right] \\
G[k] &= G[k-1] + g[k], \quad G[k-1] := 0 \quad \text{if } \Delta \bar{w}_d[k] = 0 \quad \text{or} \quad \text{sgn } \Delta \bar{w}_d[k] \neq \text{sgn } \Delta \bar{w}_d[k-1] \\
\left| \sum_{j=j_0}^k e_b[j] g[j] \right| &\leq \sqrt{P_{bb}} |G[k]| \quad (e_b \text{ one-signed on a monotone command segment } j_0..k) \\
q_{\text{law}}[k] &= P_{bb}(G[k]^2 - G[k-1]^2) = P_{bb} g[k] (2G[k-1] + g[k]) \geq 0, \quad \sum_{j=j_0}^k q_{\text{law}}[j] = P_{bb} G[k]^2
\end{aligned}$$

$$R_2 = K_{\text{var}} I F_{\text{var}} (\bar{a}_w + \xi)^2 + d \hat{b}^2 (1 - \hat{a}_w)^4 Q_{33} \quad (q_{\text{law}} \text{ not in the delay term})$$

$$\begin{aligned}
R &= \text{diag}(R_1, R_2) \\
R_1 &= \sigma_{n_w}^2 \\
R_2 &= K_{\text{var}} I F_{\text{var}} (\bar{a}_w + \xi)^2 + d Q_{44} \\
K_{\text{var}} &= \frac{2a_{\text{cov}}}{2 - a_{\text{cov}}} \\
\sigma_{\delta \bar{w}_r}^2 &= C_{dpmr} \frac{4k_B T a_o}{R} (\bar{a}_w + \xi) \\
\xi &= \frac{C_n \sigma_{n_w}^2}{C_{dpmr} (4k_B T a_o / R)}
\end{aligned}$$

Seed and prior

$$\begin{aligned}
\bar{a}(\bar{w}) &= 1 - \frac{1}{b(\bar{w} - \bar{w}_s)} \\
\left. \frac{\partial \bar{a}}{\partial b} \right|_{\bar{w}} &= \frac{1}{b^2(\bar{w} - \bar{w}_s)} = \frac{1 - \bar{a}}{b} \\
\left. \frac{\partial \bar{a}}{\partial \bar{w}_s} \right|_{\bar{w}} &= -\bar{a}'
\end{aligned}$$

$$\begin{aligned}
\hat{b}[0] &= b_{\text{true}}(\bar{a} \rightarrow 1) = \frac{8}{9} \\
P_{55}[0] &= P_{bb} = \left[\sup_{\bar{w} \in \mathcal{E}} |b_{\text{true}}(\bar{w}) - \frac{8}{9}| \right]^2 \\
\hat{a}_w[0] &= 1 - \frac{1}{\hat{b}[0] (\bar{w}[0] - \hat{w}_s)}
\end{aligned}$$

$$P_{44}[0] = (\hat{a}'[0])^2 P_{\bar{w}_s} + \left(\frac{1 - \hat{a}_w[0]}{\hat{b}[0]} \right)^2 P_{bb} + \text{floor}_a^2$$

$$P_{45}[0] = \frac{1 - \hat{a}_w[0]}{\hat{b}[0]} P_{bb}$$

$$e_{\bar{a}}(\bar{w})|_{\bar{w}_s} = -\bar{a}'(\bar{w}) e_{\bar{w}_s} \propto \frac{1}{b(\bar{w} - \bar{w}_s)^2}$$

$$e_{\bar{a}}(\bar{w})|_b = \frac{e_b}{b^2(\bar{w} - \bar{w}_s)}$$

$$e_b[k+1] - e_b[k] = b_{\text{true}}(\bar{a}_w[k+1]) - b_{\text{true}}(\bar{a}_w[k])$$

$$F_{e,55} = 1, \quad q_5 = 0 \quad (\text{consider model: the drift above is carried inside } |e_b| \leq \sqrt{P_{bb}}, \text{ not propagated})$$

$$\sum_{j=0}^{k-1} F_e[j]_{45} = \sum_{j=0}^{k-1} (1 - \hat{a}_w^+[j])^2 \left[\Delta \bar{w}_d[j] + (1 - \lambda_c) \delta \hat{w}_3[j] \right] = \frac{\hat{a}_w[k] - \hat{a}_w[0]}{\hat{b}[0]} - \frac{1}{\hat{b}[0]} \sum_{j=0}^{k-1} L_{4,\cdot}[j] \nu[j]$$

$$P_{45}[k] \approx P_{45}[0] + \frac{\hat{a}_w[k] - \hat{a}_w[0]}{\hat{b}[0]} P_{bb} = \frac{1 - \hat{a}_w[k]}{\hat{b}[0]} P_{bb}$$

$$P_{44}[k] \approx P_{44}[k]|_{\bar{w}_s, \text{floor}} + \left[\frac{1 - \hat{a}_w[k]}{\hat{b}[0]} \right]^2 P_{bb}$$

(first order: $F_{e,44} \neq 1$, the $F_{e,43}P_{35}$ coupling and the measurement updates of P_{45} , P_{44} are neglected; the second line is $e_{\bar{a}}|_b = e_b(1 - \bar{a})/b$ of the block above, held at every k instead of at $k = 0$ only)









